

National University of Computer and Emerging Sciences

Lahore Campus

Probability & Statistics

(MT2005)

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Course Instructor(s)

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MID-2 Exam

(Rubric)

Total Time (Hrs): 1

Total Marks: 45

Total Questions: 2

ALL SECTIONS BCS

CLO 2: Probability and Distributional Analysis: Apply foundational principals of probability to analyze experiments, including Bayes' theorem, evaluate discrete and continuous distributions, and explore applications in machine learning.

Q1.

Marks=[25+5]

Sol.

	free		Prize		Offer		unsubscribe		
	Yes	No	Yes	No	Yes	No	Yes	No	Total
Spam	28	12	24	16	32	8	36	4	40
Ham	16	144	8	152	20	140	10	150	160
	44	156	32	168	52	148	46	154	200

i-Email-1

free(w_1) = Yes, prize(w_2) = Yes, offer(w_3) = Yes, unsubscribe(w_4) = Yes

$P(w_1 \cap w_2 \cap w_3 \cap w_4)$

$$= P(\text{Spam}) \times P(w_1|\text{Spam}) \times P(w_2|\text{Spam}) \times P(w_3|\text{Spam}) \times P(w_4|\text{Spam}) \\ + P(\text{Ham}) \times P(w_1|\text{Ham}) \times P(w_2|\text{Ham}) \times P(w_3|\text{Ham}) \times P(w_4|\text{Ham})$$

$$P(w_1 \cap w_2 \cap w_3 \cap w_4) = \frac{40}{200} \times \frac{28}{40} \times \frac{24}{40} \times \frac{32}{40} \times \frac{36}{40} + \frac{160}{200} \times \frac{16}{160} \times \frac{8}{160} \times \frac{20}{160} \times \frac{10}{160}$$

$$= 0.06048 + 0.00003125 \Rightarrow 0.06051125$$

$$P(\text{Spam} | w_1 \cap w_2 \cap w_3 \cap w_4) = \frac{0.06048}{0.06051125} = 0.99948$$

$$P(\text{Ham} | w_1 \cap w_2 \cap w_3 \cap w_4) = \frac{0.00003125}{0.06051125} = 0.000516 \quad (\text{Also, } 1 - 0.99948 = 0.00052)$$

Since $P(\text{Spam} | w_1 \cap w_2 \cap w_3 \cap w_4) > P(\text{Ham} | w_1 \cap w_2 \cap w_3 \cap w_4)$ so Email-1 is classified as Spam.

OR

The Email-1 will be classified as spam because the probability of being it spam is 0.99948, which exceeds the threshold of 0.9.

ii- Email-2

free(w_1) = Yes, prize(w_2) = No, offer(w_3) = Yes, unsubscribe(w_4) = No

$P(w_1 \cap \bar{w}_2 \cap w_3 \cap \bar{w}_4)$

$$= P(\text{Spam}) \times P(w_1|\text{Spam}) \times P(\bar{w}_2|\text{Spam}) \times P(w_3|\text{Spam}) \times P(\bar{w}_4|\text{Spam})$$

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$$+ P(Ham) \times P(w_1|Ham) \times P(\overline{w_2}|Ham) \times P(w_3|Ham) \times P(\overline{w_4}|Ham)$$

$$P(w_1 \cap \overline{w_2} \cap w_3 \cap \overline{w_4}) = \frac{40}{200} \times \frac{28}{40} \times \frac{16}{40} \times \frac{32}{40} \times \frac{4}{40} + \frac{160}{200} \times \frac{16}{160} \times \frac{152}{160} \times \frac{20}{160} \times \frac{150}{160}$$

$$= 0.00448 + 0.00891 \Rightarrow 0.01339$$

$$P(\text{Spam} | w_1 \cap \overline{w_2} \cap w_3 \cap \overline{w_4}) = \frac{0.00448}{0.01339} = 0.334578$$

$$P(\text{Ham} | w_1 \cap \overline{w_2} \cap w_3 \cap \overline{w_4}) = \frac{0.00891}{0.01339} = 0.6654$$

Since $P(\text{Ham} | w_1 \cap \overline{w_2} \cap w_3 \cap \overline{w_4}) > P(\text{Spam} | w_1 \cap \overline{w_2} \cap w_3 \cap \overline{w_4})$ so Email-2 is classified as Ham.

OR

The Email will be classified as Ham because the probability of being it spam is 0.334578, which does not exceed the threshold of 0.9.

Marks Distribution (part a)

Identification of Prior probability (Spam and Ham) and Conditional Probabilities	Email-1 5 marks (0.5 mark for each probability)	Email-2 5 marks (0.5 mark for each probability)
	(1 mark for correct formula of total probability)	
Total Probability	Correct calculation $\Rightarrow$ 2marks	Correct calculation $\Rightarrow$ 2 marks
Posterior probability	Calculation for (Spam) $\Rightarrow$ 2marks Calculation for (Ham) $\Rightarrow$ 2marks	Calculation for (Spam) $\Rightarrow$ 2marks Calculation for (Ham) $\Rightarrow$ 2marks
Final Classification	Correct justification $\Rightarrow$ 1 mark	Correct justification $\Rightarrow$ 1 mark

Part (b): Independent Classifiers (5 Marks)

$$P(\text{at least one classifier detects as Spam}) = 1 - P(\text{no one detect})$$

$$= 1 - (0.8 \times 0.7 \times 0.5) = 0.72$$

OR

Let C_1, C_2, C_3 be the three independent classifiers

$$P(C_1 \cup C_2 \cup C_3) = P(C_1) + P(C_2) + P(C_3) - P(C_1 \cap C_2) - P(C_1 \cap C_3) - P(C_2 \cap C_3) + P(C_1 \cap C_2 \cap C_3) = 0.72$$

Marks Distribution: (b-part)

- Use of complement rule or addition rule $\rightarrow$ **2 marks**
- Correct multiplication (independence) / calculation $\rightarrow$ **2 marks**
- Final answer $\rightarrow$ **1 marks**

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Q2:

[Marks: 15]

Sr.	SCENARIOS	RESPONSE
1	A machine learning engineering team is studying the relationship between training time (X) measured in seconds and model performance (Y) measured as accuracy (in proportion) across multiple experiments. The following summary statistics are observed: $\sigma_X = 3.2$, $\sigma_Y = 0.60$, $\sigma_{XY} = 1.10$. Determine and Interpret the correlation coefficient between X and Y.	$\text{Corr}(X, Y) = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{1.10}{(3.2)(0.60)} = 0.573$ Moderate positive Correlation. [Marks: 2 (1 for formula, 0.5 for answer and 0.5 for interpretation)]
2	A system selects 2 packets without replacement from a total of N packets, where: - a = number of high-priority packets - b = number of error packets - c = number of normal packets Let 'X' and 'Y' represent number of high-priority packets selected and number of error packets selected. Construct a joint Probability Function $f(x, y)$.	$f(x, y) = \frac{\binom{a}{x} \binom{b}{y} \binom{c}{2-x-y}}{\binom{N}{2}}$ Where, $X = 0, 1, 2$ and $Y = 0, 1, 2$ $X + Y \leq 2$ [Marks: 2 (1 for function, 0.5 each range)]
3	Consider a newspaper circulation department where it is known that 84% of the households in a particular neighborhood subscribe to the daily edition of the paper. In addition, it is known that the probability that a household which already holds a daily subscription also subscribes to the Sunday edition is 0.75. What is the probability that a household subscribes to both the daily and Sunday editions of the newspaper?	$P(D \cap S) = P(D) P(S/D)$ $= (0.84)(0.75) = 0.63$ [Marks: 2 (1 formula, 1 answer)]
4	Two students scheduled interviews for summer employment at an Institute. In each case the interview results in either an offer for a position or no offer. Experimental outcomes are defined in terms of the results of the two interviews. a. List the experimental outcomes. b. Define a random variable that represents the number of offers made. Is the random variable discrete or continuous?	“O” denotes Offer, “N” denotes No Offer a- Sample space = (OO, ON, NO, NN) b- X: No of offers made, $x = 0, 1, 2$ (Discrete) [Marks: 2 (1 for a, 1 for b)]
5	A continuous RV 'X' has PDF $f(x) = e^x$, $0 < x < 1$, else 0. Check if the given function is a valid PDF or not.	A function is a valid pdf if 1) $f(x)$ is non-negative for the range of x. (true, since $e^x > 0$) 2) $\int_0^1 e^x dx = 1$ $[e^x]_0^1 = e^1 - e^0 = e - 1 \neq 1$ (Not a valid pdf) [Marks: 2 (1 for condition 1, 1 for condition 2)]
6	A random variable Y has PMF: $P(Y = 1) = 0.2$, $P(Y = 2) = 0.5$, $P(Y = 3) = 0.3$. Find the mean value of $g(x) = 2Y + 1$, if $E(Y) = 2.1$.	$E(2Y + 1) = 2 E(Y) + 1$ $= 2(2.1) + 1 = 5.2$ [Marks: 2 (1 for property, 1 for answer)]

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7	In a factory, Machine A produces 40% of the total items, of which 3% are defective. Machine B produces the remaining 60% of the items, of which 4% are defective. Are the events ‘an item is defective’ and ‘the item was produced by Machine A’ independent? Justify your answer numerically.	$P(\text{defective}) = P(D) = 0.4 \times 0.03 + 0.6 \times 0.04 = 0.012 + 0.024 = 0.036.$ $P(\text{defective from A}) = P(D/A) = 0.03.$ Since $0.03 \neq 0.036$, events are NOT independent. [Marks: 2 (1 for P(D), 1 for comparison/answer)]
8	A system generates passwords using six different characters: A, B, C, D, E, and F. Each password must use all characters exactly once, with the restriction that A and B must not appear consecutively. How many such valid passwords can be formed?	$6! - 5! \times 2! = 720 - 240 = 480$ [Marks: 1]